CO_2 Retrieval Simulator

Monte Carlo Simulation

Technical Reference & Theory Guide

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1 What Is Monte Carlo Simulation?

Monte Carlo (MC) simulation is a computational technique that uses **random sampling** to understand the statistical behaviour of a system. Rather than solving a problem analytically, we run the same experiment thousands of times with randomly drawn inputs and study the distribution of outputs.

In the context of satellite CO_2 retrieval the central question is:

Core Scientific Question

"If we observe a CO_2 scene under a given set of atmospheric conditions and instrument noise, how accurately and precisely can we retrieve the true CO_2 concentration?"

This approach mirrors the end-to-end simulator (E2ES) framework used by missions such as **OCO-2** (NASA), **MicroCarb** (CNES/UKSA), and **GOSAT** (JAXA) for pre-launch performance prediction and post-launch algorithm validation.

1.1 Why Monte Carlo for CO_2 Retrieval?

The CO_2 retrieval problem is non-linear and subject to multiple uncertainty sources that are difficult to propagate analytically:

- **Instrument noise:** photon shot noise, detector read noise, and thermal noise all add random errors to the measured spectrum.
- **Scene variability:** surface albedo, solar geometry, temperature, and pressure vary globally and affect signal strength.
- **Retrieval non-linearity:** the Beer–Lambert forward model is exponential, so errors do not propagate linearly through the inversion.
- **Prior constraint effects:** optimal estimation regularises the solution toward a prior, which can introduce systematic bias when the true CO_2 departs from the prior.

2 How the Simulation Generates Data

Each Monte Carlo trial follows a five-step pipeline:

#	Stage	What happens
1	**Parameter sampling**	Draw atmospheric and instrument parameters from probability distributions (P, T, A_s, θ_s, SNR, X_{CO_2}).
2	**Cross section**	Compute CO_2 absorption cross section $\sigma(\tilde{\nu})$ from HITRAN line data at sampled P and T.
3	**Forward model**	Simulate the noiseless satellite radiance $I(\tilde{\nu})$ via Beer–Lambert law and two-way air-mass geometry.
4	**Add noise**	Draw a Gaussian noise realisation $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and add it to $I(\tilde{\nu})$.
5	**Retrieve**	Run optimal estimation on the noisy spectrum; record $X_{CO_2}^{\text{ret}}$ and posterior uncertainty σ_{post}.

3 Step-by-Step Equations

3.1 Step 1 — Parameter Sampling

For each trial $i = 1, \ldots, N$ the following parameters are drawn independently:

Parameter	Symbol	Distribution	Range
CO_2 concentration	X_{CO_2}	$\mathcal{N}(420, 15)$ ppm	350–550 ppm
Surface albedo	A_s	$\mathcal{U}(0.05, 0.40)$	ocean → desert
Solar zenith angle	θ_s	$\mathcal{U}(10°, 70°)$	daylit scenes
Surface temperature	T_s	$\mathcal{N}(288, 12)$ K	260–310 K
Surface pressure	P_s	$\mathcal{N}(101325, 2000)$ Pa	95000–105000 Pa
Instrument SNR	SNR	$\mathcal{U}(150, 400)$	mission-dependent

Note on distribution choices

$\mathcal{N}(\mu, \sigma)$ is used for quantities that vary symmetrically around a physical mean (temperature, pressure, CO_2 concentration). $\mathcal{U}(a, b)$ is used for albedo and geometry, which span wide ranges with no preferred value.

3.2 Step 2 — Absorption Cross Section

The CO_2 absorption cross section [cm^2/molecule] is built from the HITRAN spectroscopic database as a sum of individual line contributions:

$$\sigma(\tilde{\nu}) = \sum_i S_i(T)\, \phi_i(\tilde{\nu}) \tag{1}$$

where $S_i(T)$ [cm/molecule] is the temperature-corrected line strength and $\phi_i(\tilde{\nu})$ [cm] is the normalised line profile of transition i.

3.2.1 Line Strength Temperature Correction

HITRAN tabulates line strengths at $T_{\mathrm{ref}} = 296$ K. At temperature T they are corrected by:

$$S_i(T) \;=\; S_i(T_{\mathrm{ref}}) \cdot \frac{Q(T_{\mathrm{ref}})}{Q(T)} \cdot \exp\!\left[-\frac{hc\,E''}{k_{\mathrm{B}}}\left(\frac{1}{T}-\frac{1}{T_{\mathrm{ref}}}\right)\right] \cdot \frac{1-e^{-hc\tilde{\nu}_0/k_{\mathrm{B}}T}}{1-e^{-hc\tilde{\nu}_0/k_{\mathrm{B}}T_{\mathrm{ref}}}} \tag{2}$$

- E'' [cm^{-1}] — lower-state energy of the transition
- $Q(T)$ — total internal partition function (set to 1 here as a first approximation)
- h — Planck constant (6.626×10^{-34} J s)
- c — speed of light (2.998×10^{10} cm/s)
- k_{B} — Boltzmann constant (1.381×10^{-23} J/K)

3.2.2 Voigt Line Profile

Each spectral line has a **Voigt profile** ϕ_V, the convolution of Doppler (Gaussian) and pressure (Lorentzian) broadening:

$$\phi_V(\tilde{\nu}) \;=\; \int_{-\infty}^{+\infty} \phi_D(\tilde{\nu}')\,\phi_L(\tilde{\nu}-\tilde{\nu}')\,\mathrm{d}\tilde{\nu}' \tag{3}$$

Doppler (Gaussian) broadening arises from thermal motion:

$$\phi_D(\tilde{\nu}) \;=\; \frac{\sqrt{\ln 2}}{\sqrt{\pi}\,\Delta\tilde{\nu}_D}\exp\!\left[-\ln 2\left(\frac{\tilde{\nu}-\tilde{\nu}_0}{\Delta\tilde{\nu}_D}\right)^2\right], \qquad \Delta\tilde{\nu}_D \;=\; \frac{\tilde{\nu}_0}{c}\sqrt{\frac{2k_{\mathrm{B}}T\ln 2}{m}} \tag{4}$$

where m [kg] is the mass of one CO_2 molecule.

Lorentzian (pressure) broadening arises from molecular collisions:

$$\phi_L(\tilde{\nu}) \;=\; \frac{\Delta\tilde{\nu}_L/\pi}{(\tilde{\nu}-\tilde{\nu}_0)^2+\Delta\tilde{\nu}_L^2}, \qquad \Delta\tilde{\nu}_L \;=\; \gamma_{\mathrm{air}}\left(\frac{P}{P_{\mathrm{ref}}}\right)^{n_{\mathrm{air}}}\left(\frac{T_{\mathrm{ref}}}{T}\right)^{n_{\mathrm{air}}} \tag{5}$$

Here γ_{air} [cm^{-1}/atm] is the air-broadened HWHM from HITRAN and n_{air} is the temperature exponent.

Physical regimes

At high pressure (lower atmosphere) $\Delta\tilde{\nu}_L \gg \Delta\tilde{\nu}_D$ and the profile is Lorentzian. In the upper atmosphere $\Delta\tilde{\nu}_D \gg \Delta\tilde{\nu}_L$ and it is Gaussian. The Voigt profile captures both regimes simultaneously.

3.3 Step 3 — Forward Radiance Model

The satellite observes solar photons that have travelled down through the atmosphere, reflected from the surface, and returned upward. In the shortwave infrared (1.6 μm CO_2 band), thermal emission is negligible and the physics reduces to pure absorption along the two-way path:

$$I(\tilde{\nu}) \;=\; I_{\odot}(\tilde{\nu}) \cdot A_s \cdot \cos\theta_s \cdot \exp[-M\,\sigma(\tilde{\nu})\,N_{\mathrm{col}}] \tag{6}$$

Term	Symbol	Meaning
Solar irradiance	$I_\odot(\tilde{\nu})$	Planck blackbody at $T_\odot = 5778$ K, normalised to 1
Surface albedo	A_s	Fraction of photons reflected (Lambertian surface)
Zenith weighting	$\cos\theta_s$	Fewer photons reach the surface at high solar zenith
Air-mass factor	$M = \mu_s^{-1} + \mu_v^{-1}$	Two-way geometric path enhancement
Cross section	$\sigma(\tilde{\nu})$	CO_2 absorption [cm^2/molecule]
Column amount	N_{col}	Integrated CO_2 column [molecules/cm^2]

Two-way air-mass factor:

$$M = \frac{1}{\cos\theta_{\mathrm{sun}}} + \frac{1}{\cos\theta_{\mathrm{satellite}}} \tag{7}$$

CO_2 column amount from a discretised atmosphere of L layers:

$$N_{\mathrm{col}} = \sum_{i=1}^{L} n_i \, \Delta z_i \tag{8}$$

where n_i [molecules/cm^3] is the number density and Δz_i [cm] is the thickness of layer i. Number density follows from the ideal gas law:

$$n_i = \frac{P_i \, x_{\mathrm{CO_2}}}{k_{\mathrm{B}} \, T_i} \tag{9}$$

Hydrostatic pressure profile (scale-height model, $H \approx 8.5$ km):

$$P(z) = P_0 \exp\left(-\frac{z}{H}\right) \tag{10}$$

3.4 Step 4 — Instrument Noise

In the shortwave infrared, photon shot noise dominates, well approximated by additive Gaussian noise:

$$y(\tilde{\nu}) = I(\tilde{\nu}) + \varepsilon(\tilde{\nu}), \qquad \varepsilon(\tilde{\nu}) \sim \mathcal{N}\left(0, \sigma_\varepsilon^2\right) \tag{11}$$

The noise standard deviation is tied to the instrument SNR:

$$\sigma_\varepsilon = \frac{\max_{\tilde{\nu}}\left[I(\tilde{\nu})\right]}{\mathrm{SNR}} \tag{12}$$

Each Monte Carlo trial draws an *independent* noise realisation — even for the same true scene, every trial gives a slightly different observed spectrum $y(\tilde{\nu})$.

3.5 Step 5 — Optimal Estimation Retrieval

The retrieved X_{CO_2} is obtained by inverting the forward model using the **Optimal Estimation Method** [1]. We parameterise the CO_2 column as a scalar scaling of the prior column:

$$N_{\text{col}}(\xi) \;=\; \xi\, N_{\text{col,prior}} \quad\Longrightarrow\quad X_{CO_2} \;=\; \xi\, X_{CO_2}^{\text{prior}} \tag{13}$$

3.5.1 Cost Function

The optimal estimate minimises $J(\xi)$, which balances the spectral fit against a prior constraint:

$$J(\xi) \;=\; \underbrace{\frac{\left\|\mathbf{y}-\mathbf{F}(\xi)\right\|^2}{\sigma_\varepsilon^2}}_{\text{spectral fit}} + \underbrace{\frac{(\xi-\xi_a)^2}{s_a^2}}_{\text{prior constraint}} \tag{14}$$

- $\mathbf{y}$ — noisy observed spectrum
- $\mathbf{F}(\xi)$ — forward-model spectrum at scaling ξ
- $\xi_a = 1$ — a priori scaling (prior column = reference column)
- s_a — a priori fractional uncertainty (default 0.10)
- σ_ε^2 — measurement noise variance

3.5.2 Iterative Levenberg–Marquardt Solution

Because $\mathbf{F}(\xi)$ is non-linear (exponential Beer–Lambert), the retrieval uses an iterative **Levenberg–Marquardt** (LM) solver. At iteration k:

(1) Jacobian (finite-difference approximation):

$$K_k \;=\; \frac{\partial \mathbf{F}(\xi_k)}{\partial \xi} \;\approx\; \frac{\mathbf{F}(\xi_k+\delta\xi)-\mathbf{F}(\xi_k)}{\delta\xi} \tag{15}$$

Analytically, for the Beer–Lambert model:

$$K(\tilde{\nu}) \;=\; -M\,\sigma(\tilde{\nu})\,N_{\text{col,prior}} \;\cdot\; I(\tilde{\nu}) \tag{16}$$

(2) LM update step:

$$\Delta\xi \;=\; \frac{K_k^\top \mathbf{S}_\varepsilon^{-1}(\mathbf{y}-\mathbf{F}(\xi_k)) - \mathbf{S}_a^{-1}(\xi_k-\xi_a)}{K_k^\top \mathbf{S}_\varepsilon^{-1} K_k + \mathbf{S}_a^{-1}(1+\gamma)} \tag{17}$$

$$\xi_{k+1} \;=\; \xi_k + \Delta\xi \tag{18}$$

The LM damping parameter γ is *reduced* when the cost decreases (Gauss–Newton step) and *increased* when it rises (steepest-descent step), ensuring robust convergence from any initial point.

(3) Posterior uncertainty on ξ:

$$\sigma^2_{\text{post}} = \left(K^\top \mathbf{S}_\varepsilon^{-1} K + \mathbf{S}_a^{-1}\right)^{-1} \tag{19}$$

(4) XCO_2 uncertainty in ppm:

$$\sigma_{X_{CO_2}} = \sigma_{\text{post}} \cdot X^{\text{prior}}_{CO_2} \tag{20}$$

4 Three Simulation Modes

The script runs three distinct experiments, each isolating different aspects of retrieval performance.

4.1 Mode 1 — Noise Ensemble

Scientific Question

Given a fixed atmospheric scene, how precisely can we retrieve X_{CO_2} when only instrument noise varies?

All geophysical parameters are held fixed. N independent noise realisations $\varepsilon_1, \ldots, \varepsilon_N$ are drawn and a retrieval is performed on each. The resulting ensemble $\{X^{(1)}_{CO_2}, \ldots, X^{(N)}_{CO_2}\}$ characterises the noise-driven precision.

Default configuration:

Parameter	Value	Notes
True X_{CO_2}	430 ppm	10 ppm above prior of 420 ppm
SNR	250	Typical SWIR spectrometer
Surface albedo	0.25	Vegetated land
Solar zenith angle	30°	Mid-latitude daytime
N samples	300	Adjustable via `-n_noise`

What the outputs show:

- **Error histogram:** approximately Gaussian for a well-behaved retrieval; deviations indicate non-linearity.
- **Bias μ:** mean error; should be near zero when the true state is close to the prior.
- **Precision σ:** standard deviation of errors; directly set by the instrument SNR.
- **σ_{post}:** should agree with the empirical σ if the retrieval is well-calibrated.

4.2 Mode 2 — XCO_2 Sweep

Scientific Question

Is the retrieval linear across the full range of atmospheric CO_2 concentrations expected globally?

True X_{CO_2} is swept linearly from 380 to 480 ppm in N steps. A single noisy retrieval is performed at each point with all other parameters fixed. This reveals **prior pull**: when the true state departs from the prior, the optimal estimator is biased back toward the prior.

Prior pull is quantified by the **averaging kernel** A_K:

$$X_{CO_2}^{\text{ret}} \approx A_K \, X_{CO_2}^{\text{true}} + (1 - A_K) \, X_{CO_2}^{\text{prior}} \tag{21}$$

For the scalar retrieval:

$$A_K = \frac{K^\top \mathbf{S}_\varepsilon^{-1} K \, s_a^2}{K^\top \mathbf{S}_\varepsilon^{-1} K \, s_a^2 + 1} \in [0, 1] \tag{22}$$

$A_K \to 1$ means the measurement dominates; $A_K \to 0$ means the prior dominates.

4.3 Mode 3 — Random Scene Ensemble

Scientific Question

How does the retrieval perform across the full diversity of real atmospheric scenes encountered by a satellite in one orbit?

All parameters — geophysical and instrumental — are drawn randomly for every trial. This is the most realistic test, because a satellite observes many surface types (ocean, desert, forest) at varying latitudes and seasons.

Key questions addressed:

- Does error depend on surface albedo? (Brighter surfaces give stronger signals and lower noise.)
- Does error grow at large solar zenith angles? (Longer two-way path, deeper optical depth, more non-linearity.)
- Is the posterior uncertainty σ_{post} a reliable predictor of the true error? (Critical for satellite data quality screening.)

5 Statistical Output Equations

After running N Monte Carlo trials with true values x_i and retrieved values $\hat{x}_i$:

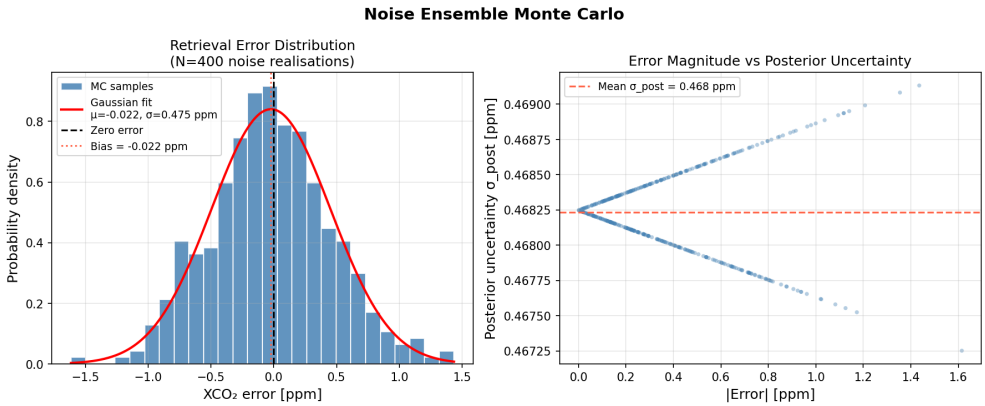

Figure 1: Noise ensemble Monte Carlo

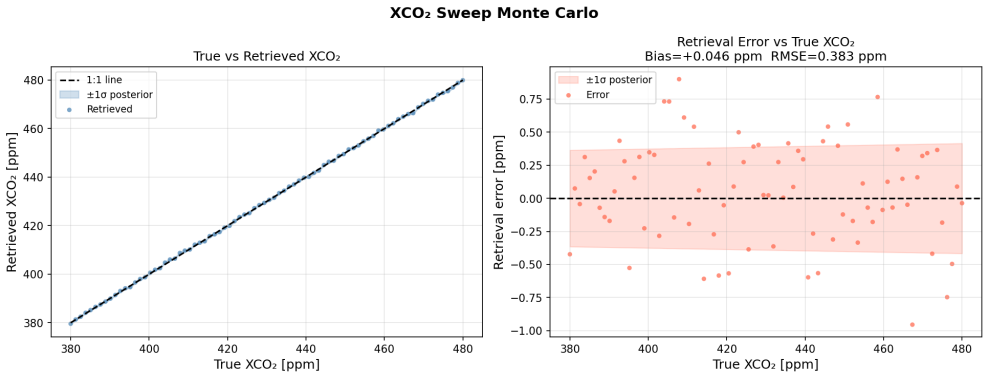

Figure 2: XCO_2 sweep

5.1 Error Metrics

$$\text{Bias} = \frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i) \tag{23}$$

$$\text{Precision } \sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\Big(\hat{x}_i - x_i - \text{Bias}\Big)^2} \tag{24}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i)^2} \tag{25}$$

These obey the **bias–variance decomposition**:

$$\text{RMSE}^2 = \text{Bias}^2 + \sigma^2 \tag{26}$$

A well-designed retrieval aims for low bias (controlled by algorithm design) and precision limited only by the instrument SNR.

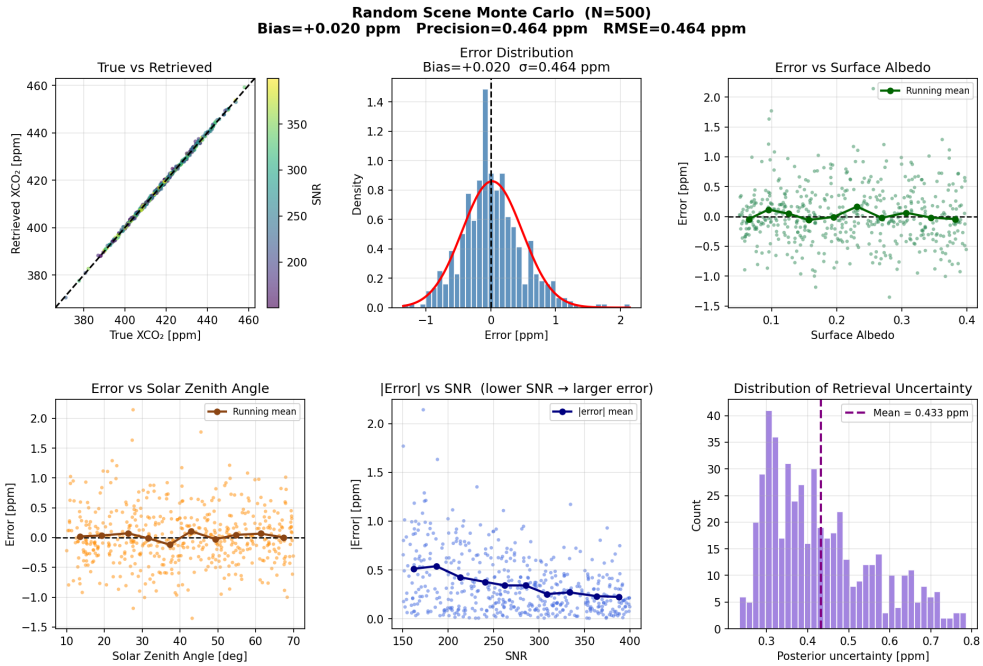

Figure 3: Random scene Monte Carlo (N=500)

5.2 SNR Scaling Law

Theory predicts that retrieval precision scales inversely with SNR:

$$\sigma_{X_{CO_2}} \propto \frac{1}{\text{SNR}} \tag{27}$$

Deviations from this 1/SNR envelope indicate non-linearity or prior constraint dominance.

5.3 Bias–Variance Trade-off

The a priori uncertainty s_a controls a fundamental trade-off:

Prior s_a	**Bias**	**Precision**	**Interpretation**
Small (tight)	High	Low	Retrieval trusts prior more than data
Large (loose)	Low	High	Retrieval trusts data more than prior
Optimal	Low	Low	Error matches prior uncertainty

6 Output Figures Guide

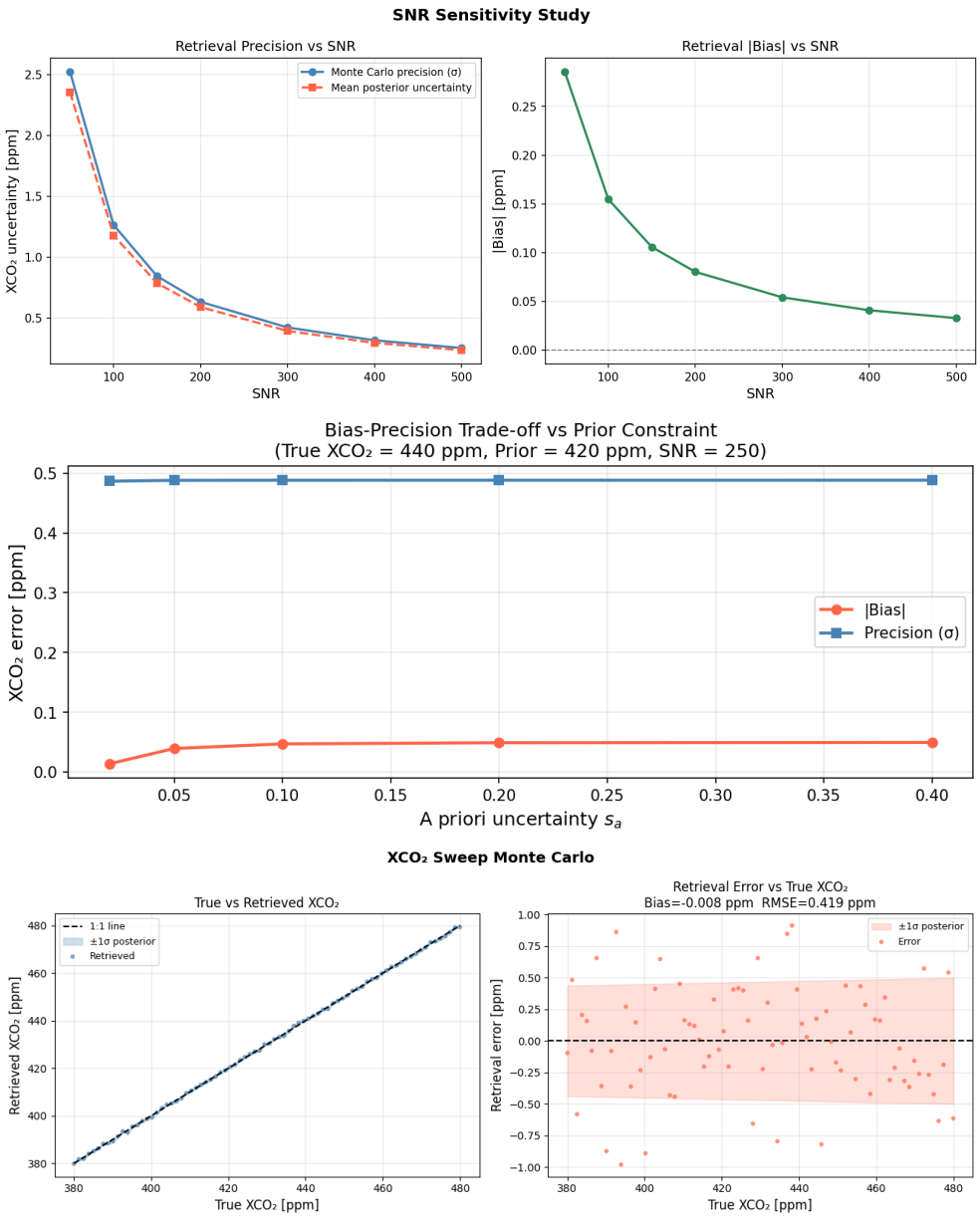

Figure 4: Fast test run 1 min ($n_noise = 100$, $n_{\text{sweep}} = 40$, $n_{\text{random}} = 150$)

Filename	Mode	Key diagnostic
`mc_01_noise_ensemble.png`	Noise Ensemble	Gaussian error histogram; $\sigma_{\text{post}} \approx \sigma$
`mc_02_xco2_sweep.png`	XCO_2 Sweep	Slope ≈ 1 in true vs. retrieved plot
`mc_03_random_scene.png`	Random Scenes	Error vs. albedo, SZA, SNR scatter panels
`mc_04_snr_sensitivity.png`	SNR Study	σ follows 1/SNR envelope
`mc_05_sza_sensitivity.png`	SZA Study	Error rise at SZA $> 60°$
`mc_06_bias_precision_summary.png`	All Modes	Side-by-side Bias, σ, RMSE

7 How to Run the Script

7.1 Requirements

```
pip install numpy scipy matplotlib
```

7.2 Default run (≈5 minutes)

```
python run_monte_carlo.py
```

7.3 Fast test run (≈1 minute)

```
python run_monte_carlo.py \
    --n_noise 100 --n_sweep 40 --n_random 150 \
    --n_snr 50 --n_sza 50
```

For this, charts are in figure-5.

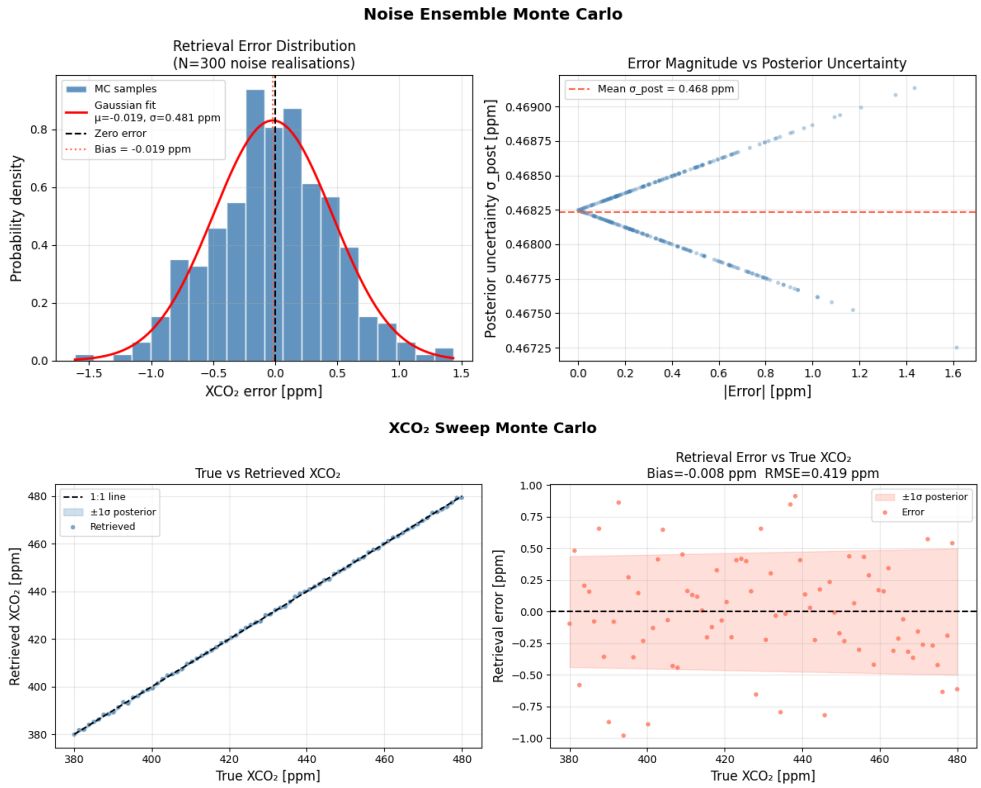

Figure 5: Fast test run 1 min ($n_noise = 100$, $n_{\text{sweep}} = 40$, $n_{\text{random}} = 150$)

Monte Carlo Retrieval Results

Two Monte Carlo experiments were performed to evaluate the performance of the CO_2 retrieval algorithm. The experiments quantify the retrieval bias, precision, root-mean-square error (RMSE), and the mean estimated uncertainty.
A noise ensemble consisting of $N = 300$ realizations was generated to evaluate the impact of measurement noise on the retrieval performance.

Noise Ensemble ($N = 300$)

- Bias: -0.019 ppm
- Precision: 0.481 ppm
- RMSE: 0.481 ppm
- Mean uncertainty: 0.468 ppm

XCO_2 Sweep ($N = 80$)

- Bias: -0.008 ppm
- Precision: 0.419 ppm
- RMSE: 0.419 ppm
- Mean uncertainty: 0.468 ppm

7.4 Headless mode (no pop-up windows)

```
python run_monte_carlo.py --no_show
```

7.5 Full command-line reference

Flag	Default	Description
`-n_noise`	300	Noise ensemble sample count
`-n_sweep`	80	XCO_2 sweep sample count
`-n_random`	400	Random scene sample count
`-n_snr`	150	Trials per SNR level
`-n_sza`	100	Trials per SZA level
`-seed`	42	Random seed (reproducibility)
`-nu_pts`	2000	Spectral grid resolution
`-no_show`	off	Suppress interactive windows

8 Physical Meaning of Each Parameter

Parameter	Symbol	Physical meaning	Why it matters
CO_2 concentration	X_{CO_2} [ppm]	Molecules of CO_2 per million dry-air molecules in the column	Controls depth of absorption lines in spectrum
Column amount	N_{col} [mol/cm^2]	Total CO_2 molecules per unit area, surface to TOA	Linear scaling of optical depth τ
Cross section	$\sigma(\tilde{\nu})$ [cm^2/mol]	Probability of photon absorption by one CO_2 molecule	Spectral fingerprint of CO_2
Optical depth	$\tau = \sigma N_{col} M$	Log-ratio of absorbed to incident photons along two-way path	$\tau > 1$: most photons absorbed
Transmittance	$\mathcal{T} = e^{-\tau}$	Fraction of photons surviving the atmospheric path	Directly measured by satellite
Air-mass factor	$M = \mu_s^{-1} + \mu_v^{-1}$	Geometric path length relative to vertical	Large SZA $\Rightarrow$ large M $\Rightarrow$ deeper absorption
Surface albedo	$A_s \in [0, 1]$	Reflectivity of the surface (Lambertian)	Ocean ($\approx$0.05) vs. desert ($\approx$0.35)
Instrument SNR	dimensionless	Signal-to-noise ratio of the spectrometer	Controls noise floor and retrieval precision
Prior s_a	dimensionless	A priori fractional uncertainty on ξ	Controls bias–variance trade-off

9 Symbol Summary

Symbol	Meaning	Units
$\tilde{\nu}$	Wavenumber	cm^{-1}
$\tilde{\nu}_0$	Line centre wavenumber	cm^{-1}
$\sigma(\tilde{\nu})$	Absorption cross section	cm^2/molecule
$S_i(T)$	Line strength at temperature T	cm/molecule
$\phi(\tilde{\nu})$	Normalised line profile	cm
$\Delta\tilde{\nu}_D$	Doppler HWHM	cm^{-1}
$\Delta\tilde{\nu}_L$	Lorentzian HWHM	cm^{-1}
E''	Lower-state energy	cm^{-1}
γ_{air}	Air-broadened HWHM at P_{ref}	cm^{-1}/atm
n_{air}	Temperature exponent	—
N_{col}	CO_2 column amount	molecules/cm^2
M	Two-way air-mass factor	—
A_s	Surface albedo	—
θ_s	Solar zenith angle	degrees
ξ	CO_2 scaling factor	—
ξ_a	A priori scaling factor	—
s_a	A priori fractional uncertainty	—
σ_ε	Noise standard deviation	radiance units
K	Jacobian $\partial F/\partial\xi$	radiance / unit ξ
A_K	Averaging kernel	—
σ_{post}	Posterior uncertainty on ξ	—
$\sigma_{X_{\mathrm{CO_2}}}$	Retrieved $X_{\mathrm{CO_2}}$ uncertainty	ppm

9 References

9 References

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